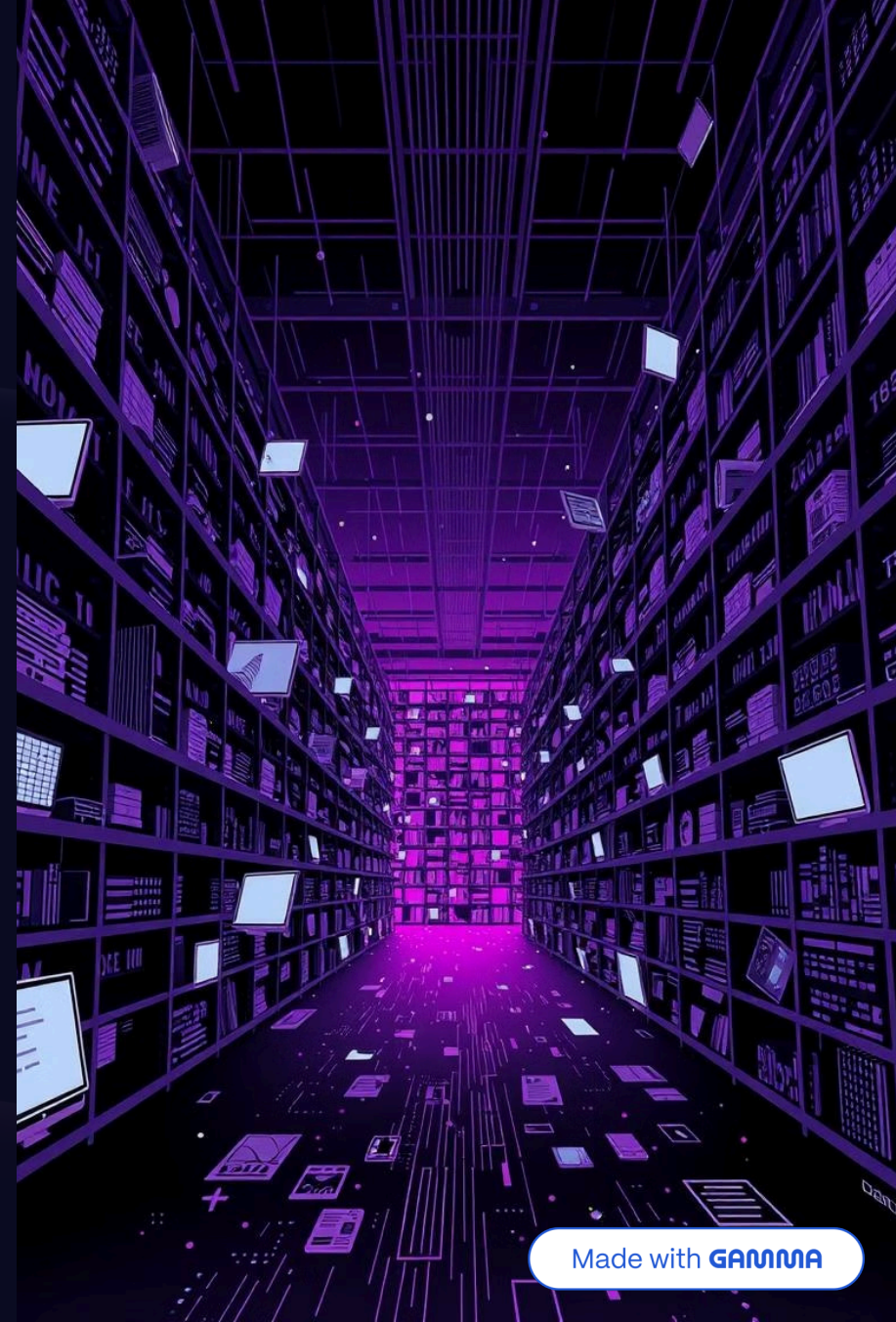


From Raw Data to Retail Insights



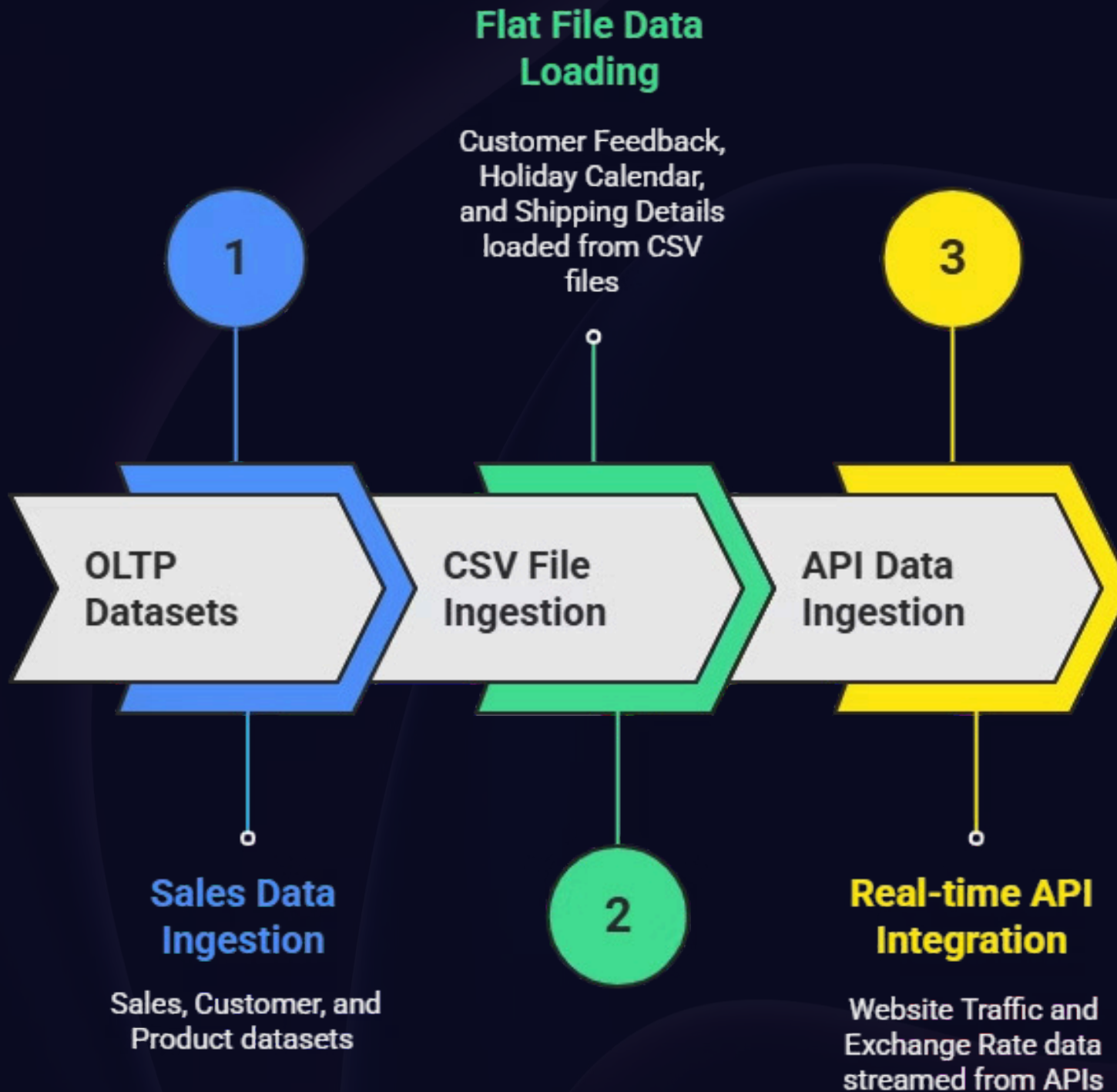
The Challenge: Scattered Retail Sales Data

Retail sales data was abundant but spread across various, disconnected sources.

- **Different databases**
- **Varied formats**
- **Inconsistent structures**
- **Lack of unified view**



Data Sources for Data Warehouse



Transforming Chaos into Clarity



Raw Data Sources

Unstructured, disparate, and inconsistent data from various origins.



Integration & Cleaning

Consolidating, validating, and standardizing data to ensure accuracy.



Structured Data Warehouse

Centralized, organized repository for efficient storage and retrieval.



Business Insights & Analytics

Leveraging refined data to drive informed decisions and strategic growth.



Our Solution: Unifying & Processing Data

We implemented a robust strategy to transform raw data into valuable business intelligence.



Gather

Consolidate diverse data sources.



Process

Clean, transform, and integrate information.



Analyze

Extract meaningful insights.

Architecting Clarity: Star Schema & Data Modeling



The Engine: ETL Pipeline Development



Extraction

Retrieving data from all source systems.



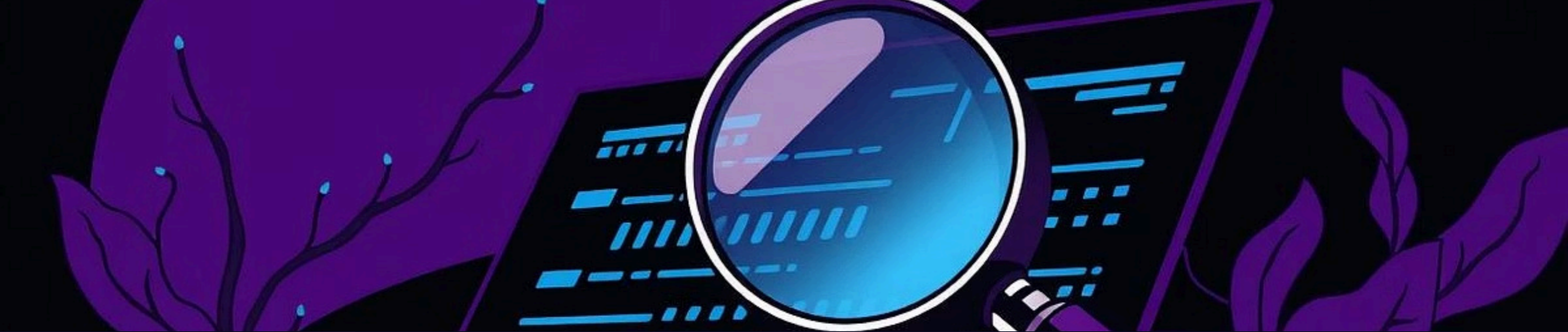
Transformation

Cleaning, standardizing, and aggregating data.



Loading

Populating the data warehouse for analysis.



Ensuring Trust: Data Quality Management

High-quality data is critical for accurate insights and reliable decision-making.



Validation

Checking for accuracy and consistency.



Cleansing

Correcting errors and removing duplicates.



Governance

Establishing policies for data integrity.

The Foundation: Data Warehouse Deployment

Building a robust and scalable environment for our consolidated data.

- Database setup and configuration
- Optimizing for query performance
- Ensuring data security and access control



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Made with **GAMMA**

Sales Analysis Dashboard

Sales Revenue

\$205.92M

Total Suppliers

13.43K

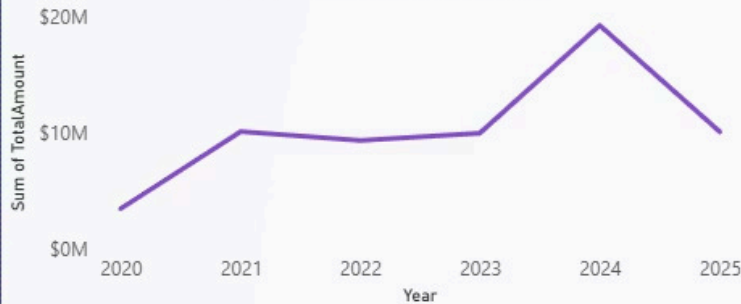
Units Sold

32M

Average Order Value

\$398

Yearly Sales Revenue Trend



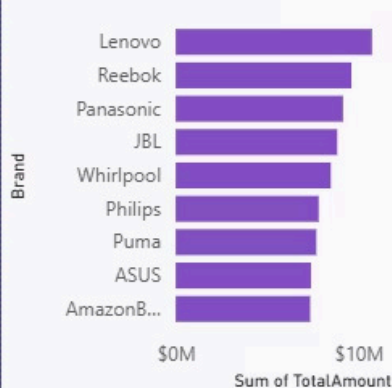
Top 3 Selling Products



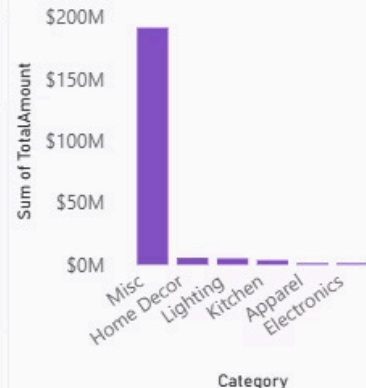
Exchange Rate By Country

Country	Currency	Exchange_GBP
United Arab Emirates	AED	0.21
Australia	AUD	0.50
Bahrain	BHD	2.00
Brazil	BRL	0.14
Canada	CAD	0.54
Switzerland	CHF	0.95
Czech Republic	CZK	0.04

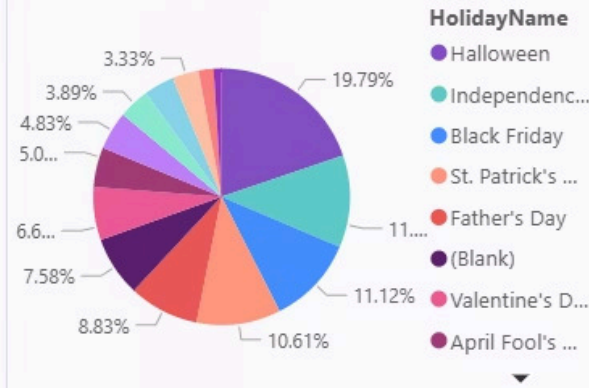
Sales Revenue by Brand



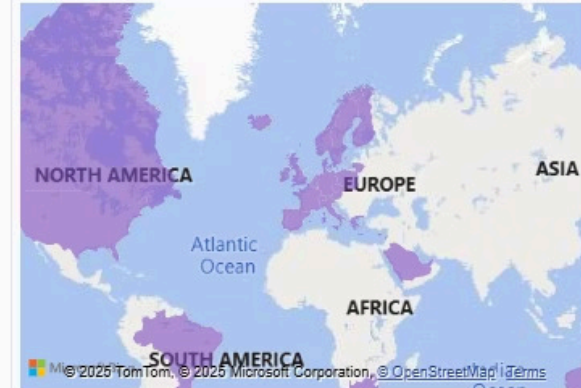
Sales Revenue by Category



Average Holiday Sales Revenue



Sales Revenue by Country



Customers Analysis Dashboard

Sales Revenue

\$205.92M

Total Customers

4360

Total Transactions

19.77K

Average Delivery Days

4

Page Views by Country

United Kingdom

32191708

Germany

867464

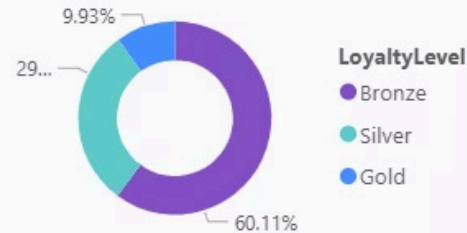
France

692553

EIRE

555923

Customer Segmentation



Monthly Website Traffic Trend



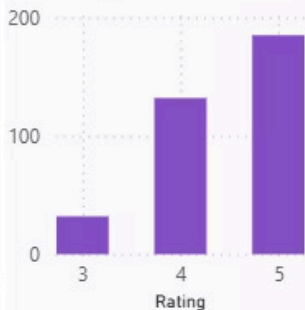
Top 10 Customers (CLV)

CustomerKey	CustomerName
1	Terri Acosta
16	Frank Young
67	George Rose
1313	Anne Kelly
2138	Mark Smith
3009	Wendy King MD
3277	Briana Jones
3447	Shelby Kennedy
3726	James Coleman

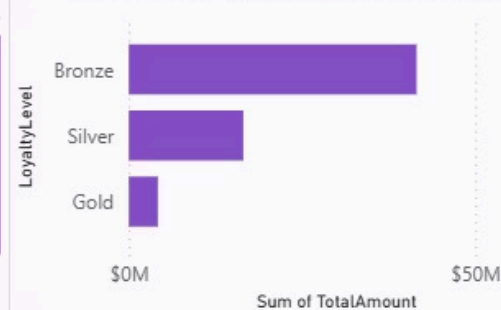
Customer Acquisition Trend



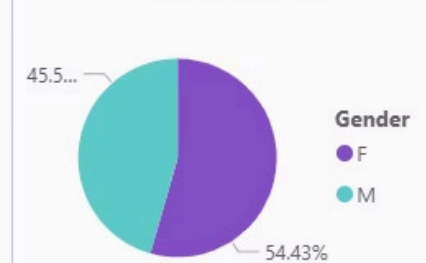
Rating Distribution



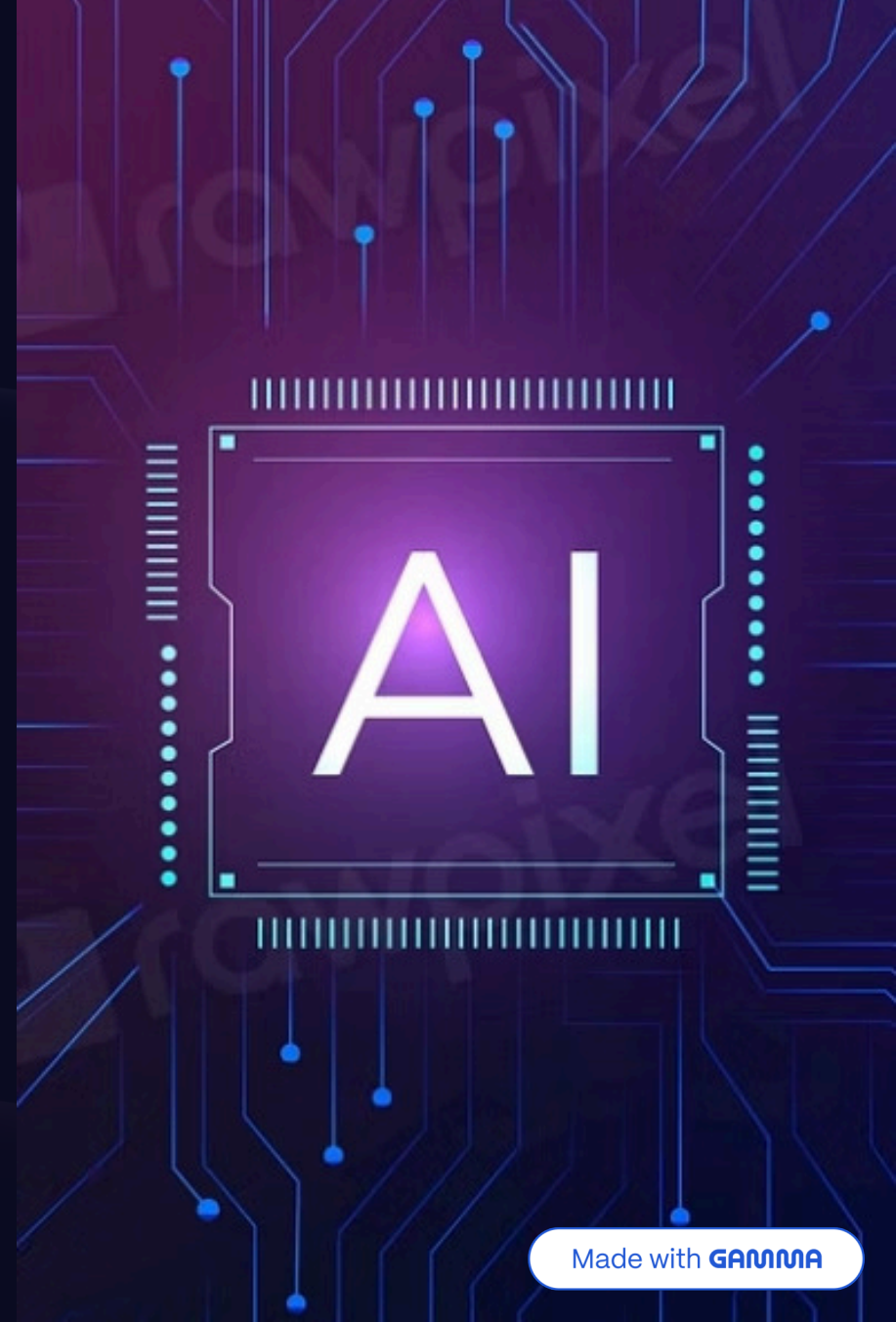
Revenue by Customer Loyalty Level



Sales by Gender

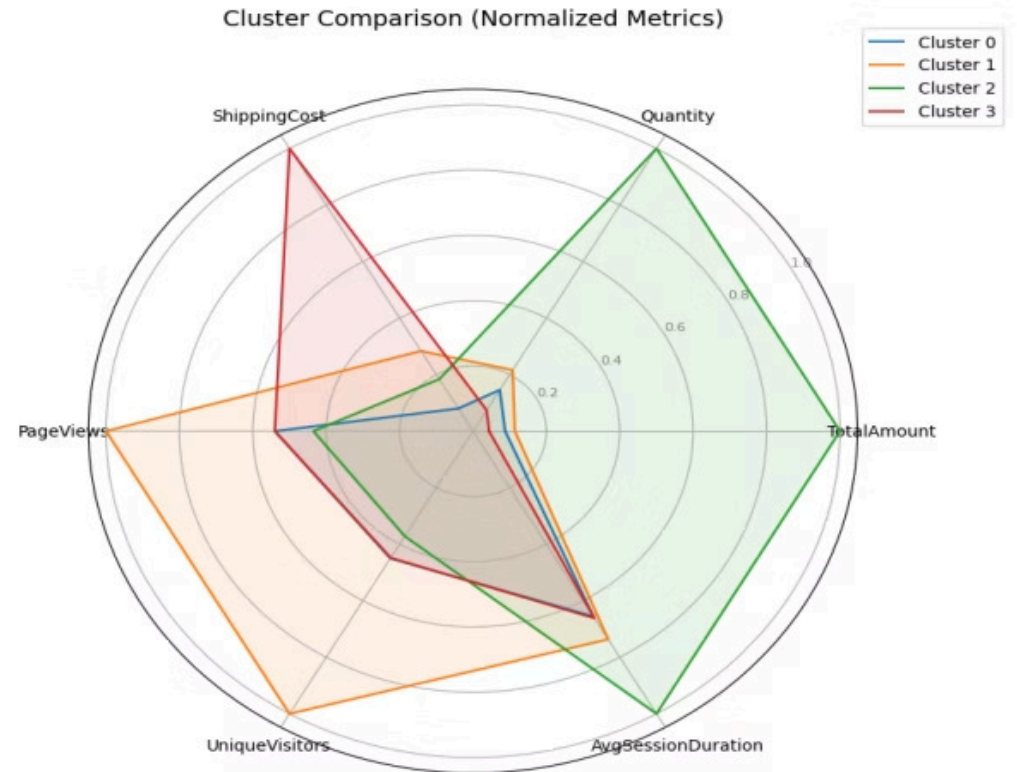


Analytics & ML Insights



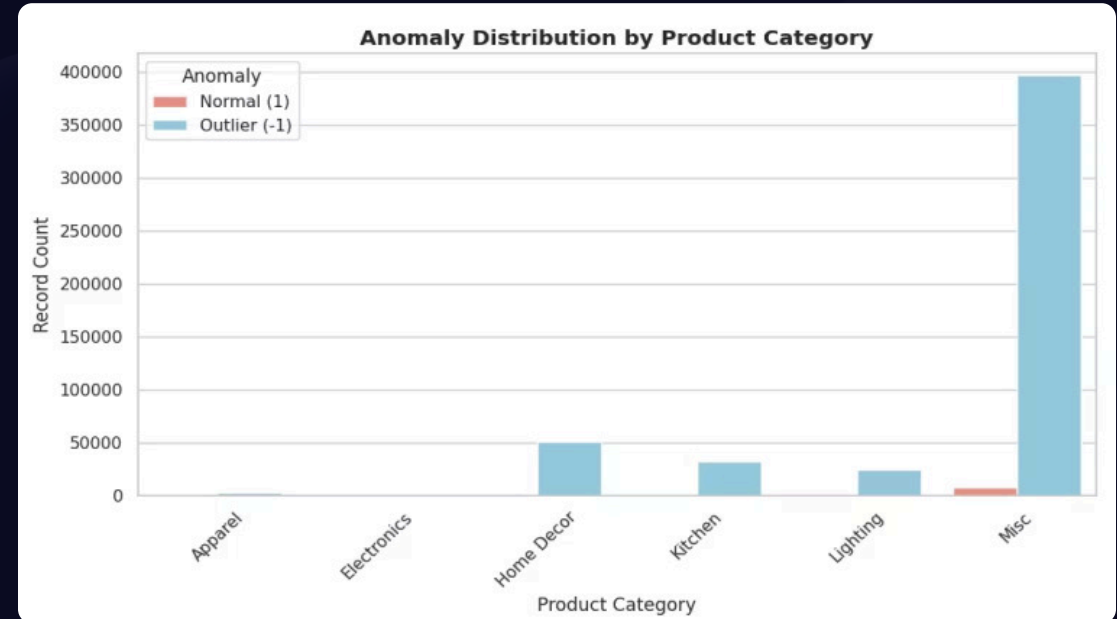
K-Means Clustering

- **Customer Segmentation:** Employed K-Means clustering to group customers by purchase frequency and transaction value.
- **Insights Gained:** Identified key segments such as high-value and low-engagement shoppers.
- **Impact:** Enabled targeted marketing and optimized inventory management.



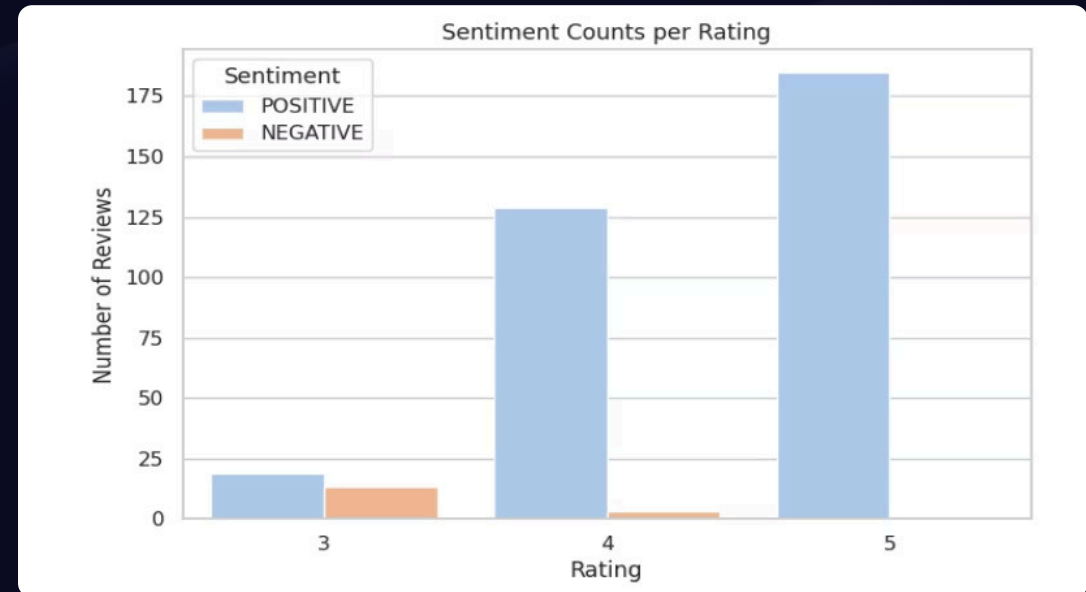
Anomaly Detection

- **Isolation Forest Model:** Applied machine learning to detect anomalies in sales transaction data.
- **Use Cases:** Identified data entry errors, fraud indicators, and unusual purchasing behaviors.
- **Outcome:** Improved data reliability and supported proactive fraud prevention.



Sentiment Analysis

- **Transformer-based Sentiment Analysis:** Used pretrained models (e.g., BERT) to classify customer feedback as positive, neutral, or negative.
- **Actionable Insights:** Helped identify key improvement areas and track satisfaction trends over time.
- **Business Impact:** Enabled data-driven decisions for better products, services, and customer experiences.



Sentiment Counts per Rating

Any Questions?